

RYAN GREEN, PH.D.

Los Angeles, CA | rygreen@ucsc.edu | ryanagreen.github.io | LinkedIn: ryangreenphd | GitHub: RyanAGreen

SUMMARY

Ocean and climate scientist with 7+ years of quantitative research and industry experience, translating science into credible MRV and life-cycle accounting systems. Lead MRV for a first-of-a-kind mCDR demonstration plant, spanning carbon accounting and LCA, scientific and process data management, environmental monitoring, and third-party validation and verification.

EXPERIENCE

Ocean Systems Lead | Equatic

May 2025 – Present

- Built Equatic's MRV framework and Project Design Document for a first-of-a-kind mCDR demonstration plant, translating carbon-registry protocol requirements into facility measurement, reporting, and third-party validation and verification requirements.
- Built the data management plan and infrastructure for our demonstration plant, connecting continuous process (SCADA) data, laboratory analytical results, and environmental monitoring data into a single auditable MRV pipeline.
- Leading life-cycle assessment and GHG-accounting development for Equatic's carbon-removal process; developed an internal LCA model and associated emissions-factor library (GREET, ICE, DESNZ, supplier data) and reconcile outputs against Isometric and Puro.earth protocols.
- Leading environmental monitoring and permitting efforts for Equatic's current and future plants, working directly with regulators, laboratories, and third-party verification bodies.
- Leading ecotoxicology and environmental risk studies for Equatic's discharge, including bioassay design, feedstock characterization, and near-field plume modeling across multiple facility configurations.

Marine CDR Consultant | Oceanid MRV / Oceanid Climate and Carbon Solutions

2023 – 2025

- Advised on ocean carbon cycling and carbonate chemistry across multiple marine CDR methods and clients; helped develop MRV frameworks for a range of mCDR pathways.

Graduate Student Researcher | University of California, Santa Cruz

2019 – 2025

- Developed and applied a coupled regional-global carbon-cycle model in Python to study geologic analogs of ocean alkalinity enhancement in the Eastern Tropical North Pacific.
- Evaluated the use of carbon isotopes as MRV tools for marine CDR, and simulated ocean alkalinity enhancement and direct ocean capture deployment scenarios within the California Current System using the Regional Ocean Modeling System (ROMS).

EDUCATION

Ph.D., Earth and Ocean Sciences | University of California, Santa Cruz

2019 – 2025

Dissertation: "Carbon Isotopes as Tools for Understanding Natural and Engineered Ocean Alkalinity Enhancement." Extensive Python-based scientific computing with regional and global ocean-carbon models.

B.S., Environmental Policy (Minor: Oceanography) | University of California, Davis

2013 – 2018

Graduated with honors and a departmental citation.

SELECTED PUBLICATIONS

Corsini, L., **Green, R.A.**, Nguyen, T.T.M., et al. (2026) *Ecotoxicological evaluation of effluents from an electrolytic marine carbon dioxide removal process*. Submitted.

Green, R.A., Rafter, P.A., Sun, C., et al. (2026). *Simulating deglacial radiocarbon anomalies with pH-neutral geologic carbon*. *Paleoceanography and Paleoclimatology*, 41, e2025PA005217.

Green, R.A., Hain, M.P., & Rafter, P.A. (2024). *Deglacial pulse of neutralized carbon from the Pacific seafloor: A natural analog for ocean alkalinity enhancement?* *Geophysical Research Letters*, 51, e2024GL108271.

TECHNICAL SKILLS

Python (scientific computing & data analysis) • LCA & emissions-factor evaluation (openLCA, GREET, ICE, DESNZ) • GHG accounting & carbon-removal MRV • Isometric / Puro.earth protocol interpretation & compliance • Data architecture (SCADA, LIMS, cloud data warehousing) • Carbonate chemistry & ocean/carbon-cycle modeling (ROMS) • Cross-functional translation of science into operational and technical specifications • Environmental permitting